

Week 4 Progress Writeup

1. Each team member's contribution:

John Won - This week I focused on scoping Model 1, the campaign response probability model. I reviewed Claudia's Data Agent research to align the Model 1 design with the shared data pipeline, and compiled slides for our mentors covering the business problem, the relevant tables, campaign types, and data volume. I defined the prediction setup as one row per household-campaign mailing (7,208 rows, 889 positives), with the label set to at least one redemption during the campaign window and all features restricted to data before the campaign's start day to prevent leakage. I summarized Model 1 in the mentors' six-question format and drafted a recommendation on which agents can be reused across the three model objectives. During our meeting with mentors on Friday, I presented the Model 1 scoping and design sections. I also worked on crafting the slides for the Problem Description presentation slated for this Monday.

Zhiyan Yang - This week I focused on planning and validating the Data Agent, the first agent in our pipeline. I read through the dunnhumby user guide and our project proposal and wrote a detailed guideline for whoever builds the agent. It covers how to process each of the 8 tables, why, and what the outputs should look like so that they serve all three prediction questions, not just the first. I also wrote a shorter summary for the rest of the team. To check my assumptions against the real data, I wrote a verification script and ran it on all 8 raw files. It reproduced the proposal's numbers exactly (7,208 household-campaign mailings, 889 with a redemption, 12.33%), and it surfaced several issues. The proposal's net revenue formula would add the coupon discount back instead of subtracting it. For TypeA campaigns we know only the pool of coupons (about 180 to 210), not the 16 each household received. Campaigns overlap heavily (74 of 435 pairs) and are not in date order, so train/test splits need to go by date. There are also conflicting duplicate rows in the display/mailed table, and week_no follows $\text{ceil}((\text{day} + 2) / 7)$ rather than the obvious $\text{ceil}(\text{day} / 7)$. I also built a first-draft ingestion script that loads, validates and cleans all 8 files, including the 36.8-million-row display/mailed table. It is a loader only, and the cross-table checks and the household $\times$ campaign table are still to be built. Along with that, I set up a repo structure and README for the team and prepared slides 1 to 5 for our mentor meeting.

Shreya Shetty - This week I fully designed and led the development strategy for Model 3 (True Incremental Uplift), successfully pivoting our approach from a basic time-series comparison to a robust causal inference framework and researched more into its feasibility with our data. I proved the dataset's feasibility by engineering the complex joins between the transaction logs and `causal\_data` for a tiny test set, allowing us to use a T-Learner methodology to mathematically isolate the true impact of direct-mail campaigns from in-store confounders like promotional displays. To mitigate risks prior to ML training, I built a diagnostic Python script to validate control group volume and seamlessly translate daily campaign windows into weekly transaction data, and I translated this entire architectural upgrade into slides for Model 3 in our group presentation deck that I presented during our mentor meeting to secure buy-in on the new methodology. In addition to owning Model 3, I collaborated closely with Anna (who was leading Model 2 design) to refine the `SALES\_VALUE` schema logic for Model 2 (Expected Financial Value), successfully coordinated with the team to

schedule our internal and follow-up meetings to keep our development momentum on track, and also worked on the Problem Description Slides for our Capstone Class presentation.

Pranav Jain - This week I designed the agent architecture for our system. I went through each agent one by one (Data, Feature, Modeling, Evaluation, Scoring, Decision Lab and Decision Support) and worked out what each one takes in, what it produces and what it decides, for each of our three models. I concluded that one shared architecture works for all three models: each agent does the same job in the same way, and only the instructions it receives change with the model. For example, the Feature Agent first builds one table of 75,000 household $\times$ campaign rows containing every model's features and targets, and then each model keeps only the campaigns, rows and target it needs (Model 3 keeps only TypeB/C campaigns). The Modeling Agent first reshapes the data for each model type and then trains it, and the Evaluation Agent tests every model on later campaigns against a simple baseline. Model 3 plugs its own two-model method and Qini evaluation into these shared agents. To ground these decisions, I profiled all 8 raw tables myself. Among other things, I found that zero-dollar receipt lines carry 24% of all coupon discounts, which is why the Data Agent should flag unusual rows rather than delete them. I wrote this up in the Agent Architecture document in our team folder, built the agent architecture slides, and presented them during our mentor meeting.

Yuxin Cai (Anna) - This week I worked on researching and scoping Model 2 and translating the initial proposal into a more concrete modeling approach. I reviewed the dataset structure, did preliminary eda, and refined the target definition, including confirming that SALES_VALUE should be used directly rather than subtracting the retailer coupon-match discount again. I defined Model 2 as a household-campaign level regression estimating expected financial value conditional on redemption. I also investigated how to define campaign-related product value and identified a key limitation for Type A campaigns: each household receives 16 personalized coupons, but the specific coupons are not observed in the dataset. Since Type A represents a large share of redeemed pairs, I began exploring how to handle this limitation while retaining these campaigns. I discussed the Model 2 design and its distinction from Model 3's causal objective with Shreya, compiled the research and open questions into slides for our mentor meeting, and contributed to the Problem Description presentation for our Capstone class.

2. Plan for upcoming week

Next week we plan to build out the first two agents properly. For the Data Agent, that means moving the first-draft loader into the repo, adding the cross-table checks, building the household $\times$ campaign table and producing the data quality report. For the Feature Agent, it means building the shared feature table and the step that selects each model's own campaigns, rows and target. In parallel, we will research Model 1 in depth so that we know exactly how to build it once the first two agents are ready. This includes deciding which household features to build and how far back before each campaign to look, which models to try first (for example logistic regression and gradient boosting), and how to handle the fact that only about 12% of mailings lead to a redemption. It also includes choosing where to split earlier and later campaigns for training and testing, how to treat households without demographics, and how to measure success: the share of redeemers captured in the top 20% of scores, compared with simple baselines such as random selection and targeting households that redeemed before.